

Roll No.: .....

**END SEMESTER EXAMINATION DECEMBER 2025**

Name of the Program: B.Tech

Semester: V

Name of the Course: Cloud-Based Application Development & Management Course Code: TCS552

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

| Q1  | (20Marks)                                                                                                                                                                            |     |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| (a) | Explain the role of cloud data centers. Discuss the issue and challenges of designing data centers.                                                                                  | CO1 |
| (b) | Explain the role and taxonomy of SLA's for the fulfillment of cloud based services.                                                                                                  | CO1 |
| (c) | Explain the architecture of Software-Defined Networking (SDN) and how the Software-Defined Networking (SDN) framework is integrated and applied within cloud computing environments. | CO1 |
| Q2  |                                                                                                                                                                                      |     |
| (a) | Explain the concept of Sandboxing. Discuss the steps of application development and deployment in Google App engine platform.                                                        | CO2 |
| (b) | Examine the concept of buckets. How resource naming is done in AWS storage services?                                                                                                 | CO3 |
| (c) | Compare and contrast the cloud services provided by Amazon ElastiCache and Amazon CloudFront.                                                                                        | CO4 |
| Q3  |                                                                                                                                                                                      |     |
| (a) | Illustrate the goals of energy efficiency in clouds and briefly explain the green cloud computing architecture with a suitable diagram.                                              | CO3 |
| (b) | Explain the concept of Cloud federation along with various technologies used in cloud federation.                                                                                    | CO3 |
| (c) | Explain the key role of virtualization and how the CPU, memory and I/O resources are virtualized.                                                                                    | CO3 |
| Q4  |                                                                                                                                                                                      |     |
| (a) | Discuss the requirements, issues and challenges in Cloud management activities.                                                                                                      | CO5 |
| (b) | Discuss the significance of cloud policy and governance in ensuring effective risk management and compliance with regulatory practices.                                              | CO5 |
| (c) | Discuss any case study describing it's use case related to Dynamo DB and Cassandra.                                                                                                  | CO5 |
| Q5  |                                                                                                                                                                                      |     |
| (a) | Explain the impact of IoT, Big Data, and Machine Learning on cloud computing. Discuss briefly about map reduce programming model.                                                    | CO5 |
| (b) | Frame how application-level and host-level security are implemented in cloud computing environments. Why cloud computing security is required at this levels ?                       | CO6 |
| (c) | Discuss any case study describing it's use case related to Azure Digital Twins and AWS IoT TwinMaker.                                                                                | CO6 |

